

GARV PUNDIR

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Software Engineer — AI/ML & Data, with hands-on experience across backend systems, data pipelines, and applied machine learning.

EDUCATION

University of Wisconsin – Madison <i>B.S. Computer Science, Data Science, and Economics Expected December 2026</i> Dean's List: Spring 2023, Spring 2025 GPA: 3.6/4.0 Relevant Coursework: Artificial Intelligence, Data Structures & Algorithms, Machine Organization, Database Management Systems, Software Engineering, Object-Oriented Programming, UI/UX Design	Madison, WI
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TECHNICAL SKILLS

Languages: Java, Python, C, C++, SQL, JavaScript, TypeScript, HTML/CSS, R
AI / ML: PyTorch, CUDA, Scikit-learn, LLM API integration (OpenAI, Anthropic), RAG pipelines, LangChain, vector databases (FAISS/Chroma), LoRA/PEFT fine-tuning, NumPy, pandas
Frameworks: React, Node.js, Flask, FastAPI, Spring Boot, JUnit, Docker
Databases: PostgreSQL, MySQL, SQLite, NoSQL
Tools: Git, Agile Development, Linux, Bash, Vim, IntelliJ, Visual Studio

WORK EXPERIENCE

University of Wisconsin – Madison — Undergraduate Research Assistant <i>Tabular Synthetic Data Generation & Model Benchmarking (TabDLM, TabDiff, CTGAN) PyTorch, CUDA, Docker, LoRA</i> - Benchmarked 3 state-of-the-art generative AI architectures (CTGAN, TabDiff, an 8B-parameter LLM) across GAN, diffusion, and LLM paradigms, evaluating fidelity, downstream utility, and privacy-leakage risk on mixed-type security datasets. - Synthesized high-fidelity tabular records spanning numerical, categorical, and unstructured-text attributes by fine-tuning an 8B-parameter foundation model (TabDLM) with LoRA. - Accelerated inference throughput 3.5x — cutting generation time from 21 hours to ~6 hours for 5,500+ records — by diagnosing CUDA out-of-memory errors and optimizing batch size and sampling-step depth on 48GB GPUs. - Standardized reproducible benchmarking across all 3 model paradigms by containerizing environments in Docker/Conda and automating schema-validation scripts for synthetic output.	June 2026 – Present
University of Wisconsin – Madison, Dept. of Computer Sciences — Peer Mentor, CS354 - Selected as 1 of 5 peer mentors for CS354 (250 students) and mentored ~50 students weekly on low-level systems programming — virtual memory, stack/heap allocation, pointer arithmetic, cache organization, and x86 assembly. - Guided students in analyzing cache behavior and performance tradeoffs — spatial/temporal locality, cache-miss patterns, and optimization techniques in C — to strengthen their systems-level debugging skills.	Jan 2025 – May 2025
Harmonizer Global — Software Development Engineering Intern <i>Remote, India</i> - Shipped real-time temperature and precipitation visualization widgets across 4 IoT device types by integrating REST-based device APIs to fetch and render live sensor data. - Cut client-side rendering latency ~40% by refactoring asynchronous data handling and eliminating unnecessary DOM reflows across web and embedded interfaces. - Restored widget stability across heterogeneous IoT hardware by identifying and resolving memory leaks and performance bottlenecks through structured debugging and browser profiling.	Aug 2024 – Oct 2024

PROJECTS

LLM-Based Course Alignment Python, Flask, SQL Automated course- and lecture-level instructional-quality analysis for faculty by building a Flask backend with relational SQL schemas that used LLMs to validate alignment between learning objectives and course materials (piloted on CS320).	Sept 2025 – Dec 2025
LLM Grading Quality Research Python, NLP - Improved LLM-generated grading consistency and feedback specificity by building Python evaluation pipelines that benchmarked model output against rubrics for structure, coherence, argument strength, and grammar. - Fine-tuned prompts and model parameters to improve grading performance and consistency, producing more actionable, specific feedback for students.	Sept 2024 – Jan 2025
TimeCoin React, Java, Spring Boot, PostgreSQL, JWT <i>Team project (CS 506) — 6-person team, served as Product Owner github.com/garvpundir01/TimeCoin</i> - Delivered 25.5 story points by building the mining/rewards feature — cooldown timers, eligibility logic, and a scheduled service that batch-mints coins into the ledger. - Built the marketplace listing engine (CRUD, search/browse) and secured all API endpoints with JWT + 2FA authentication and backend input validation, syncing listing completions to real-time transaction/balance updates.	Jan 2026 – May 2026